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Exercise 2: Knowledge Graph Extraction from Text

Task 1: Natural Language Processing (NLP)

Given is the following sentence: *Barack Obama visited Berlin, the capital of Germany.*

Task 1.1 Apply the following methods on the sentence: tokenisation, lemmatisation, POS tagging (with simple tags: *noun*, *verb*, *punctuation*, *other*).

Tokens										
Lemmas										
POS										

Task 1.2 Use the Stanford CoreNLP demo[1] and show the outputs of the following methods: *lemmas*, *parts-of-speech*, *dependency parse*.

Task 2: Named Entity Recognition (NER)

Given is the sentence of Task 1.

Task 2.1 Which named entities appear in the sentence? What is their type (*Person*, *Organisation*, *Location*, *Other*)?

Entity	Type

Entity	Type

Task 2.2, 2.3 & 2.4:

➢ Jupyter Notebook *Exercise2.ipynb*, Task 2.

[1] Stanford CoreNLP demo: https://corenlp.run/

Task 3: Named Entity Linking (NEL)

Given is the following sentence: *Obama and Bush meet in Chicago.*
Within this sentence, the following mentions and candidate entities are found, together with their links and mention-entity popularity prior scores:

Mention	**Entity (abbreviation)**	**Links**	**Popularity Prior**
Obama	Barack Obama (BO)	USA, US President, Malia Ann Obama, Hillary Clinton	$\frac{4}{5}$
	Michelle Obama (MO)	USA, Malia Ann Obama, Princeton University	$\frac{1}{5}$
Bush	George H. W. Bush (GHWB)	USA, Bill Clinton	$\frac{1}{2}$
	George W. Bush (GWB)	USA, Hillary Clinton	$\frac{1}{2}$
Chicago	Chicago (C)	USA, Illinois, Lake Michigan	$\frac{9}{10}$
	Chicago (band) (CB)	Rock, Jazz, Robert Lamm	$\frac{1}{10}$

Task 3.1 Compute the link overlaps for all required entity-entity pairs.

	BO	MO	GHWB	GWB	C	CB
BO						
MO						
GHWB						
GWB						
C						
CB						

Task 3.2 Use the simplified NEL approach introduced in the lecture to find the correct entities per mention.

Use the following equation (with $\alpha = \beta = 1$) for computing the scores for the single mention m_i and an entity e_i given a set of candidate entities $C(m_j)$ per mention m_j, where $rel(e_i, e_j)$ is the link overlap between entities e_i and e_{ij}:

$$score(m_i, e_i) = \alpha \cdot prior(m_i, e_i) + \beta \cdot \sum_{m_j \in M, i \neq j} \sum_{e_j \in C(m_j),} rel(e_i, e_j)$$

Step	Mention-Entity	Score	Delete?
1	Obama-BO		
	Obama-MO		
	Bush-GHWB		
	Bush-GWB		
	Chicago-C		
	Chicago-CB		
2			
3			

Result:

Mention	Entity
Obama	
Bush	
Chicago	

Task 4: Relation Extraction (RE)

Given is the following text (with underlined entity mentions):
Obama and Bush meet in Chicago. They participated in the G7 summit which happened in April 2012.

Task 4.1 Which relations can you detect (use your own properties)?

Task 4.2 Which challenges do you see? How could you solve them?